Portugal's General-Government Balance by Subsector, 1977–2025

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Abstract

This report decomposes Portugal's annual general-government net lending (+) / net borrowing (-) into Central Government, Regional and Local Government, and Social Security Funds (SSF), for the 49 years from 1977 to 2025. The analysis is empirical and accounting-focused. It covers long-run balance composition, exact year-to-year attribution, revenue and expenditure dynamics, sign persistence, conservative structural mean shifts, Social Security revenue composition and internal systems, primary balances and interest, public investment and debt-flow reconciliation.

The central accounting identity is

$$B_t^{GG} = B_t^{C} + B_t^{RL} + B_t^{SSF},$$

and it closes for every year in the panel to within rounding.

The recorded pattern is the following. The aggregate balance is positive in 4 of 49 years: 2019, 2023, 2024 and 2025. The Central Government balance is negative in all 49, and the Social Security balance is positive in 43. In each of the 4 years with a positive aggregate balance the combined non-SSF balance is negative and the SSF balance exceeds it in absolute size. Magnitudes differ sharply across the 1995 statistical splice – the aggregate balance averages -6.43% of GDP before it and -4.04% after – so no level statistic is pooled across it. Once interest is excluded, the Central Government primary balance is positive in 15 of the 45 years for which the detailed accounts exist: a persistently negative B.9 does not imply a persistently negative primary balance.

No causal, normative, or policy-intent interpretation is assigned to any result. Every figure and table in this report is generated from a persisted artefact listed in Appendix E; no value is transcribed by hand.

Contents

1 Results at a Glance

Table 1: Headline quantities, each read from a persisted artefact.

Quantity	Value
Panel coverage	1977–2025
Annual observations	49
Sector-year detailed account observations	184
Years with a positive aggregate balance	4
General Government balance, 2025	2,059 M EUR (0.67% GDP)
Combined non-SSF balance, 2025	-5,006 M EUR (-1.63% GDP)
Social Security Funds balance, 2025	7,065 M EUR (2.30% GDP)
Maximum absolute subsector closure residual	1.00 M EUR
Maximum absolute account identity error	5.46e-12 M EUR
Maximum absolute debt reconciliation residual	0.00e+00 M EUR
Subsector account years left un-imputed	4 (1996–1999)

Residual quantities are validation checks on the extraction and the accounting identities, not economic results.

The three residual rows are the report's own audit trail. The subsector identity closes to within the one-million-euro rounding of the published sources; revenue minus expenditure reproduces the recorded balance to numerical precision; and the modern debt change reconciles exactly with the balance and the stock-flow adjustment. A change in any of those rows would indicate a defect in extraction rather than a finding.

2 Data, Sources and Validation

The canonical balance panel takes 1977–1994 from the Banco de Portugal / INE historical long series and 1995–2025 from INE data distributed through PORDATA. The CFP ESA 2010 workbooks supply modern revenue, expenditure, interest, investment, Maastricht debt and stock-flow data, and act as an independent cross-check on the balance bridge. Every workbook is parsed programmatically from the bundled file; no value is transcribed.

Table 2: Bundled sources, the coverage used from each, and the first twelve hexadecimal digits of the SHA-256 digest of the retained file.

Source	Institution	Coverage used	Vintage	SHA-256 prefix
Banco Portugal Long Series	Banco de Portugal / INE	1977-1995 for the historical segment used here	2023-12	d3570ea90f78
PORDATA Balance By Level	INE via PORDATA	1995-2025	2026-04	2a5d54649041
CFP Annual General Government	Portuguese Public Finance Council (CFP)	1995-2025	2026-04-15	8f6b5a2916a4
CFP Annual Subsectors	Portuguese Public Finance Council (CFP)	2000-2025	2026-04-15	4d9e1b07cd38
Eurostat Subsector Balances	Eurostat	1995-2025, EU and EFTA reporters, B9 by subsector	2026-07-21	cf4c9408c237
INE Wage Bill	Statistics Portugal (INE) via Eurostat	1995-2025, total economy	2026-08-17	e925c41100d8
INE Employment	Statistics Portugal (INE) via Eurostat	1995-2025, total economy	2026-08-17	7595dfe29633
CFP Social Security Detail	Portuguese Public Finance Council (CFP)	2019-2025 systems; detailed 2024-2025 tables	2026-04	d9f28e424ab6

Download URLs and local paths are recorded in `config/sources.yml`; full digests for every bundled file are in `outputs/metrics/raw_file_sha256.json`.

2.1 Data vintage

Every figure in this report belongs to one data vintage, recorded per source in Table 2. The modern national-accounts and CFP files are the April 2026 releases. The 2024 and 2025 observations are flagged **provisional** by the publisher and are carried with that flag through the canonical panel. They are the years this report discusses most, and they are the years most likely to be restated: a later vintage will revise them.

2.2 Two statistical regimes, not one series

The statistical splice at **1995** is retained explicitly and nothing is smoothed, chained or calibrated across it. Both 1995 vintages are kept, and the revision between them is reported in Table 3. The revisions are small in level terms but non-zero for every subsector, which is why no model in this report is fitted across the boundary, and why no statistic that depends on the level of the balance is averaged across it.

Table 3: The 1995 overlap. Both vintages are retained; the canonical panel uses the modern one.

Balance	Historical vintage (M EUR)	Modern vintage (M EUR)	Revision (M EUR)
General Government	-4,557.7	-4,611.0	-53.3
Central Government	-4,220.7	-4,235.0	-14.3
Regional and Local	32.6	66.0	33.4
Social Security Funds	-369.5	-442.0	-72.5

2.3 An explicit gap, not an interpolation

Detailed revenue and expenditure components are available for General Government from 1977 onward without interruption, but for the three subsectors only for 1977–1995 and 2000–2025. The four intervening years are absent from the sources and are therefore absent here; Figure 1 shows the gap directly. The canonical B.9 balance panel is unaffected: it has an observation for every one of the 49 years, and only the detailed components are missing. Every figure drawn from the account panel breaks its line across those years rather than joining 1995 to 2000.

Table 4: Coverage of the detailed account panel. The absent years are the 1996–1999 subsector components, which are missing from both sources and are not imputed.

Sector	First year	Last year	Observations	Years absent
General Government	1977	2025	49	0
Central Government	1977	2025	45	4
Regional and Local	1977	2025	45	4
Social Security Funds	1977	2025	45	4

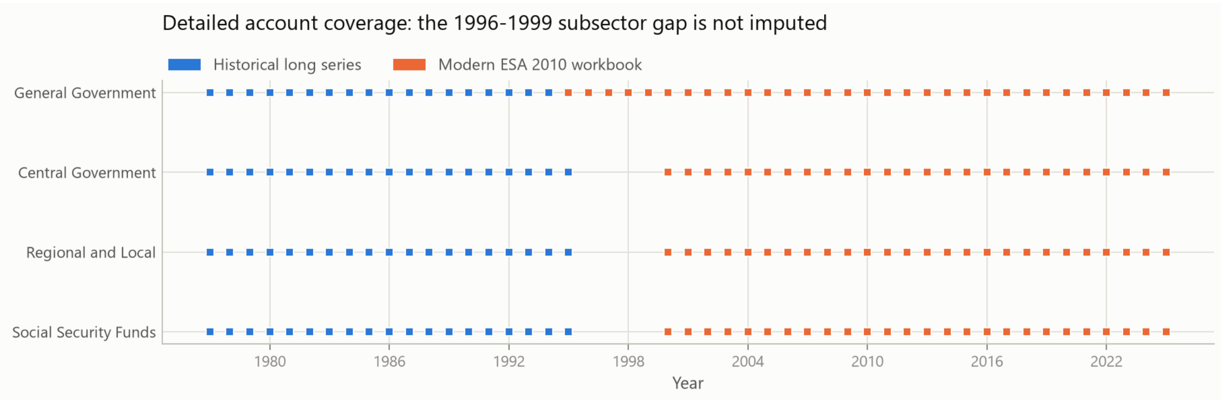

Figure 1: Detailed account coverage by sector and statistical regime. Each mark is a sector-year for which revenue and expenditure components exist. The empty band at 1996–1999 is the source gap, not a rendering artefact. Sources: Banco de Portugal / INE long series and CFP ESA 2010 workbooks.

2.4 Identity closure and source agreement are different tests

Table 5: Every cross-check the pipeline runs, in one unit. Identity rows test whether the extraction is arithmetically self-consistent; source-agreement rows test whether two independently published sources report the same number for the same year.

Check	Quantity compared	N	Largest absolute difference (M EUR)	Where
Accounting identity	Aggregate balance minus the sum of its three subsectors	49	1.000	2004
Accounting identity	Revenue minus expenditure minus the recorded balance	184	0.000	Central Government, 1994
Accounting identity	Debt change against minus the balance plus the stock-flow adjustment	108	0.000	2000
Source agreement	PORDATA bridge against the CFP workbook, General Government	31	0.498	1995
Source agreement	PORDATA bridge against the CFP workbook, Central Government	26	66.811	2002
Source agreement	PORDATA bridge against the CFP workbook, Regional and Local	26	0.492	2002
Source agreement	PORDATA bridge against the CFP workbook, Social Security Funds	26	0.470	2019
Vintage revision	1995 modern vintage against the historical vintage, four balances	4	72.465	1995

Identity closure does not imply source agreement: the identities close to rounding while the largest source disagreement is a Central Government difference of about 67 million euro. The two are different tests and neither substitutes for the other.

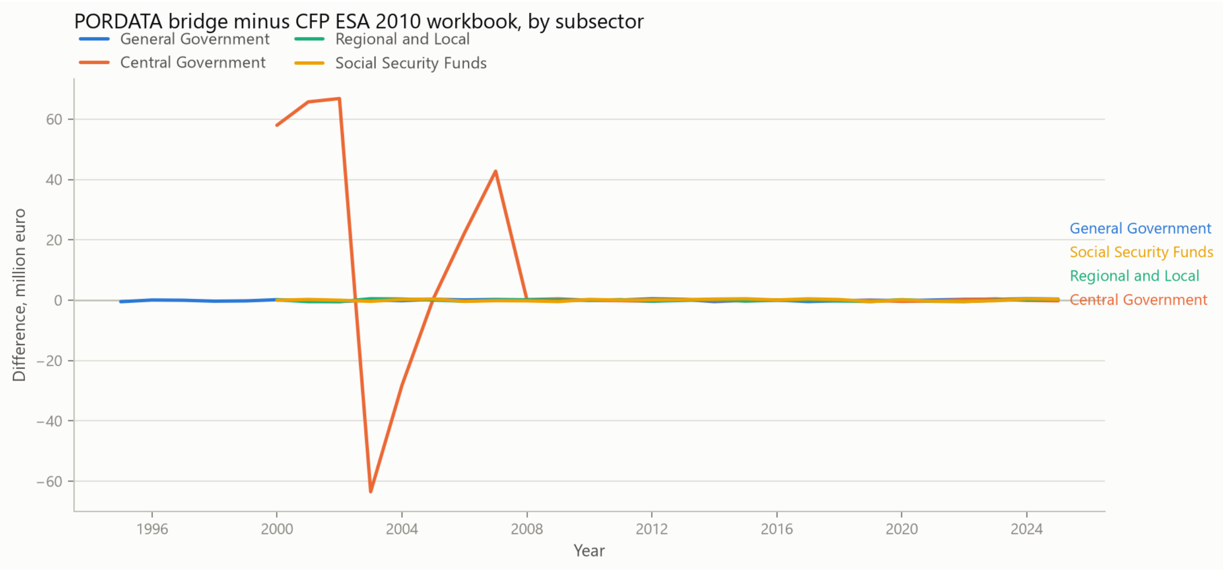

Figure 2: PORDATA bridge minus CFP ESA 2010 workbook, by subsector, over the years both sources cover. Values are million euro; zero means the two sources agree exactly. The Central Government series is the only one that departs materially from zero.

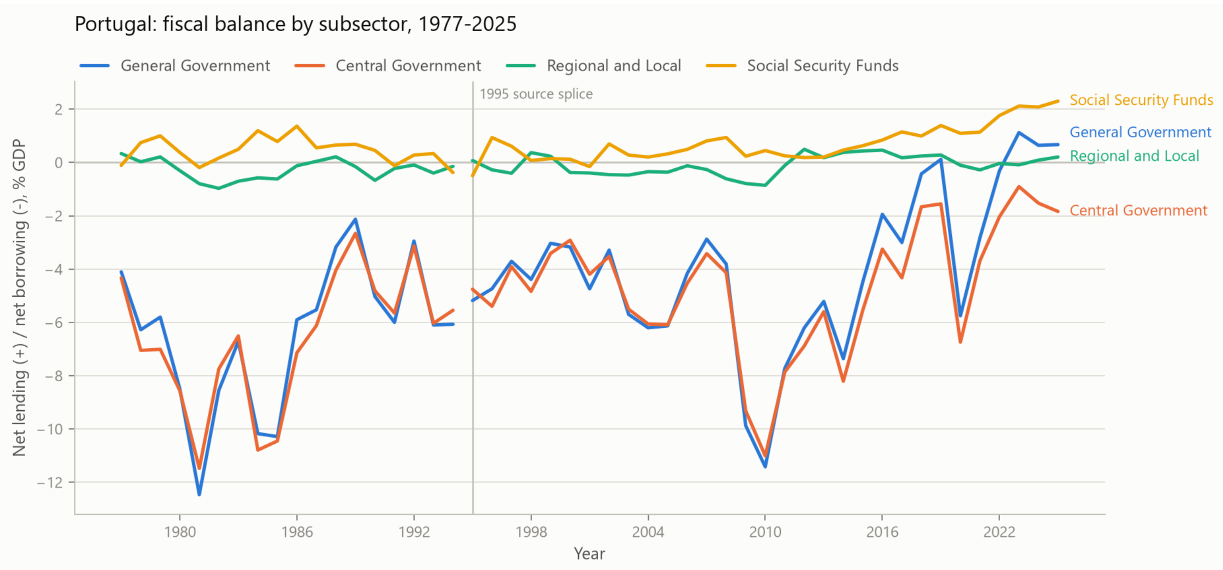

Figure 3: Net lending (+) / net borrowing (-) by subsector, as a share of GDP. The vertical rule marks the 1995 source splice.

3 Long-Run Subsector Decomposition

Over the full panel the aggregate balance is positive in 4 years: **2019, 2023, 2024 and 2025**. The Central Government balance is negative in **49 of 49 observations**, while the Social Security balance is positive in **43 observations** (Table 23 reports the full sign frequencies). These are descriptive frequencies. They are not counterfactual statements about what the aggregate balance would have been under a different institutional arrangement.

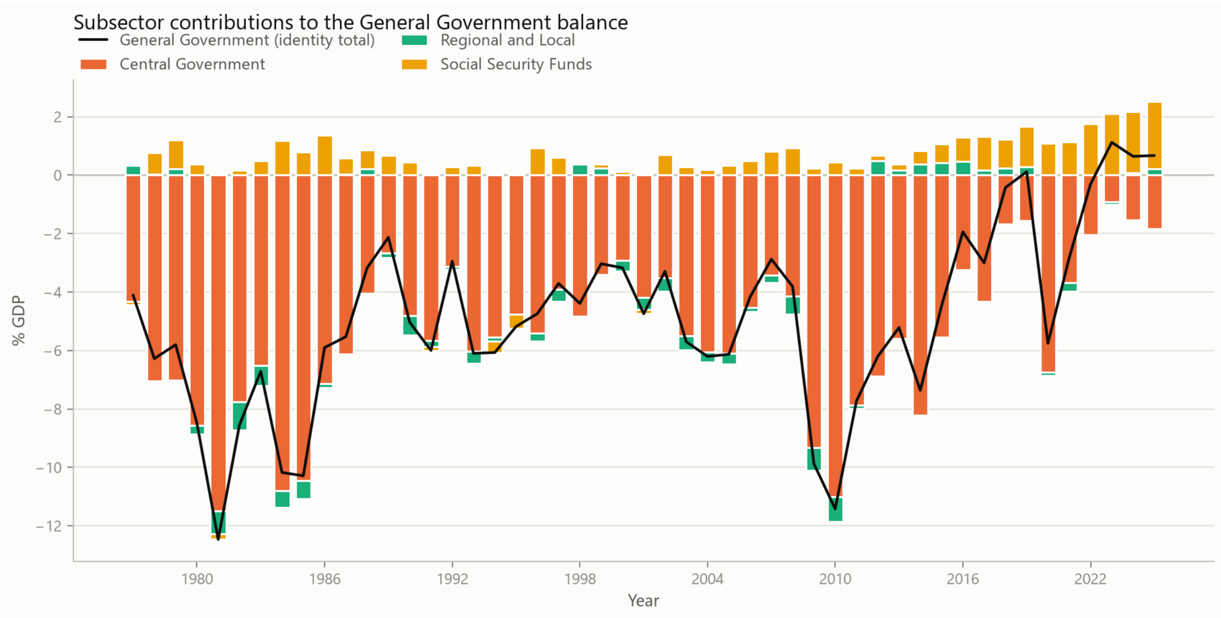

Figure 4: The identity drawn: positive and negative subsector contributions are stacked separately from zero, and the General Government line gives their algebraic sum. The total visible span of a column is therefore the sum of the absolute contributions, not the aggregate balance; a tall span under a shallow line means the subsectors offset one another that year.

3.1 Offset metrics

Writing the combined non-Social-Security balance as $B_t^{nonSSF} = B_t^C + B_t^{RL}$, the offset ratio is defined, whenever $B_t^{nonSSF} < 0$ and $B_t^{SSF} > 0$, as

$$O_t = \frac{B_t^{SSF}}{|B_t^{nonSSF}|}.$$

$O_t = 1$ means the positive Social Security balance is exactly the size of the negative non-SSF balance. The ratio compares two recorded numbers. It is not a counterfactual, and it does not describe what either balance would be under different institutional arrangements. It is undefined, and left missing, whenever the signs do not make the comparison meaningful.

In 2025 the identity reads

$$2,059 = -5,636 + 630 + 7,065 \quad \text{M EUR},$$

with a combined non-SSF balance of **-5,006 M EUR** and an offset ratio of **1.411**.

4 Year-to-Year Attribution

The annual change in the aggregate balance decomposes exactly:

$$\Delta B_t^{GG} = \Delta B_t^C + \Delta B_t^{RL} + \Delta B_t^{SSF}.$$

Table 6: Years with a positive aggregate balance.

Year	GG (M EUR)	Central (M EUR)	Regional/local (M EUR)	SSF (M EUR)	Non-SSF (M EUR)	Offset ratio
2019	249	-3,331	604	2,975	-2,727	1.091
2023	3,030	-2,447	-243	5,720	-2,690	2.126
2024	1,863	-4,424	256	6,031	-4,168	1.447
2025	2,059	-5,636	630	7,065	-5,006	1.411

Table 7: Subsector decomposition of the recent period.

Year	GG (M EUR)	Central (M EUR)	Regional/local (M EUR)	SSF (M EUR)	Non-SSF (M EUR)	Offset ratio
2010	-20,537	-19,799	-1,543	805	-21,342	0.038
2011	-13,624	-13,862	-201	439	-14,063	0.031
2012	-10,453	-11,603	835	314	-10,768	0.029
2013	-8,898	-9,555	309	348	-9,246	0.038
2014	-12,752	-14,214	655	808	-13,559	0.060
2015	-8,026	-9,931	776	1,129	-9,155	0.123
2016	-3,622	-6,055	863	1,570	-5,192	0.302
2017	-5,870	-8,460	348	2,243	-8,112	0.277
2018	-873	-3,407	499	2,035	-2,908	0.700
2019	249	-3,331	604	2,975	-2,727	1.091
2020	-11,564	-13,555	-207	2,198	-13,762	0.160
2021	-6,117	-7,985	-601	2,468	-8,586	0.287
2022	-757	-4,966	-91	4,299	-5,057	0.850
2023	3,030	-2,447	-243	5,720	-2,690	2.126
2024	1,863	-4,424	256	6,031	-4,168	1.447
2025	2,059	-5,636	630	7,065	-5,006	1.411

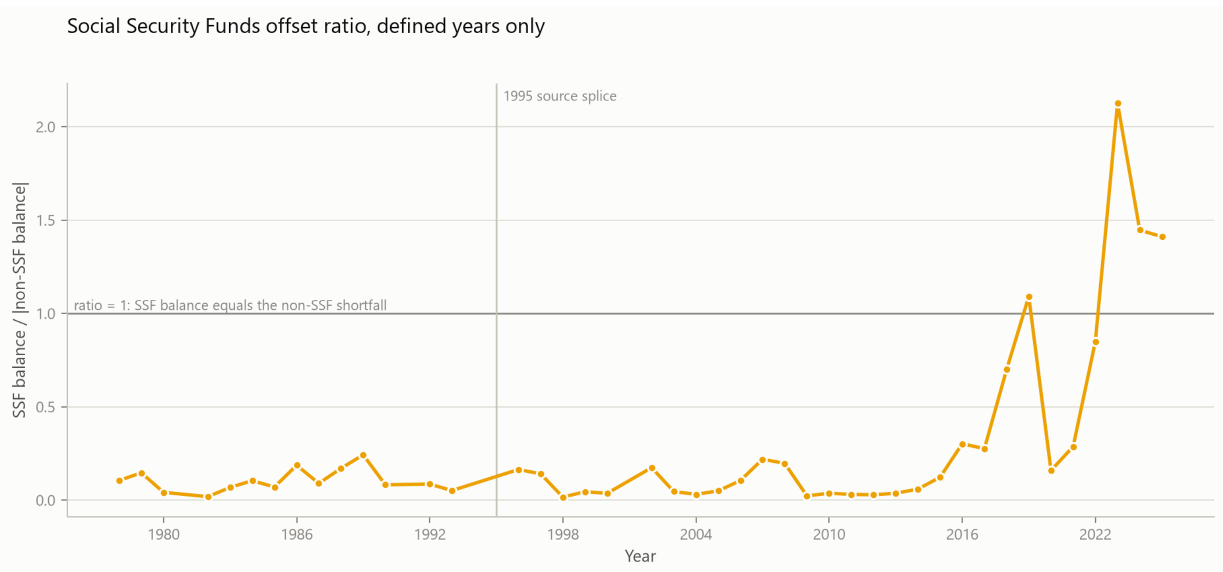

Figure 5: Social Security Funds offset ratio for the years in which it is defined.

The decomposition is computed for all **48 adjacent-year pairs** in the panel and closes to **1.00 M EUR** at worst. It separates the *level* of a subsector balance from its *contribution* to an annual improvement or deterioration, which are distinct questions that the level series alone cannot answer.

Table 8: The eight largest annual movements in the General Government balance by nominal size, and how they decompose across subsectors.

Year	Aggregate change (M EUR)	Central (M EUR)	Regional/local (M EUR)	SSF (M EUR)
2009	-10,496	-8,939	-289	-1,268
2011	6,913	5,937	1,342	-366
2015	4,726	4,283	121	321
2016	4,404	3,876	87	441
2018	4,997	5,053	151	-208
2020	-11,813	-10,224	-811	-777
2021	5,447	5,570	-394	270
2022	5,360	3,019	510	1,831

Ranked by the absolute nominal change, so the modern period dominates: nominal GDP in 2025 is orders of magnitude larger than in 1977. Table 9 ranks the same quantity scaled by GDP, which does not have that bias.

Table 9: The largest annual movements inside each statistical regime, ranked by the absolute change scaled by current-year GDP, with the subsector that accounts for most of each move.

Regime	Rank	Year	Direction	Aggregate change (% GDP)	Aggregate change (M EUR)	Central (M EUR)	Regional/local (M EUR)	SSF (M EUR)	Largest contributor	Its share of the move
1977–1994 historical	1	1981	Deterioration	-5.40	-478	-385	-48	-45	Central Government	-0.81
1977–1994 historical	2	1984	Deterioration	-4.71	-790	-922	-1	133	Central Government	-1.17
1977–1994 historical	3	1980	Deterioration	-4.02	-296	-233	-34	-29	Central Government	-0.79
1977–1994 historical	4	1993	Deterioration	-3.28	-2,222	-2,057	-210	46	Central Government	-0.93
1977–1994 historical	5	1990	Deterioration	-3.26	-1,635	-1,305	-271	-58	Central Government	-0.80
1995–2025 modern	1	2009	Deterioration	-5.98	-10,496	-8,939	-289	-1,268	Central Government	-0.85
1995–2025 modern	2	2020	Deterioration	-5.88	-11,813	-10,224	-811	-777	Central Government	-0.87
1995–2025 modern	3	2011	Improvement	3.92	6,913	5,937	1,342	-366	Central Government	0.86
1995–2025 modern	4	2015	Improvement	2.63	4,726	4,283	121	321	Central Government	0.91
1995–2025 modern	5	2021	Improvement	2.52	5,447	5,570	-394	270	Central Government	1.02

Ranking is within a regime, not across both: each annual change is computed inside one source family, but ordering historical against modern episodes by size would compare two methodologies. Scaling by current-year GDP shares one denominator across the three subsector terms, so the decomposition stays exact; it is not the change in the balance ratio, which would also move with the denominator. 1995 is excluded because the 1994-to-1995 change straddles the splice in both panels.

Both windows of the decomposition are drawn below. Figure 6 covers 1996–2025 and Figure 7 covers 1978–1994; neither crosses the 1995 splice.

Contribution is not causation. A subsector accounting for most of an annual improvement has not been shown to have produced it.

5 Revenue and Expenditure Dynamics

For every sector-year with detailed components the balance is an identity, $B_{i,t} = R_{i,t} - E_{i,t}$, and therefore so is its change,

$$\Delta B_{i,t} = \Delta R_{i,t} - \Delta E_{i,t}.$$

The decomposition reproduces the recorded balance change to **5.5e-12 M EUR** across **177 sector-year changes**. Source gaps are never bridged: no change is computed across the 1995-to-2000 discontinuity in subsector components, so those rows are dropped rather than interpolated.

5.1 Which accounts moved

The subsector split says where an episode sits; the component split says which accounts produced it. The two source families resolve the accounts at different depths, so each sector-year uses the finer scheme it reports rather than being coarsened to a common one.

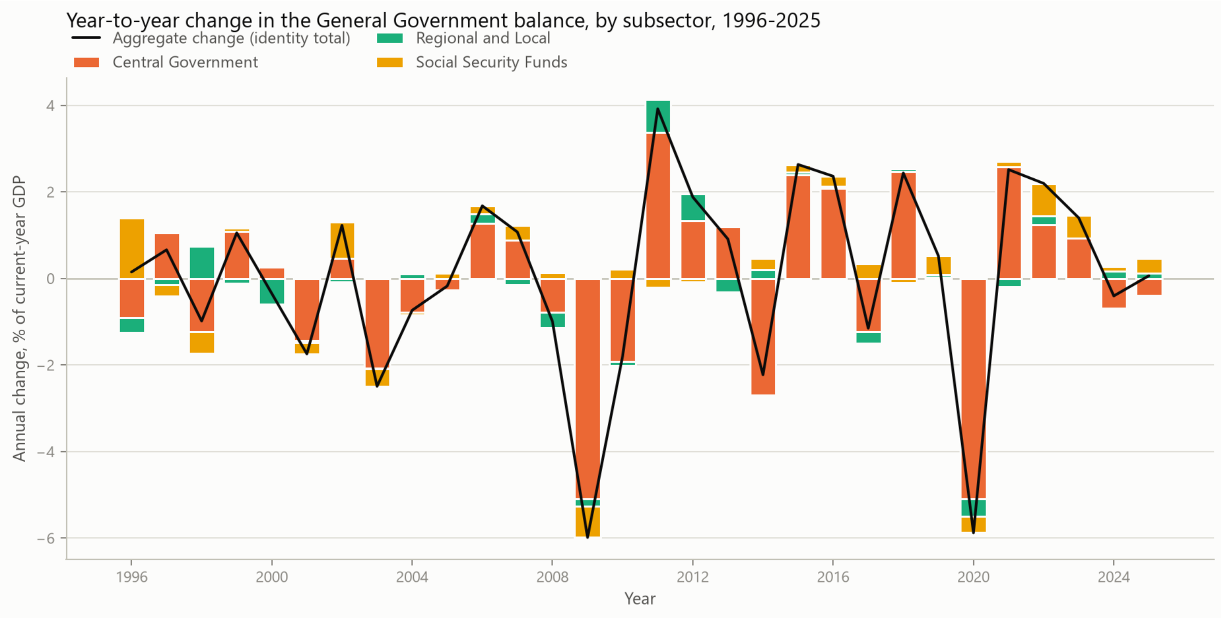

Figure 6: Annual change in the General Government balance attributed to subsectors, 1996–2025, as a percentage of current-year GDP. The black line is the identity total; a tall stack under a short line means the subsectors moved in opposite directions that year.

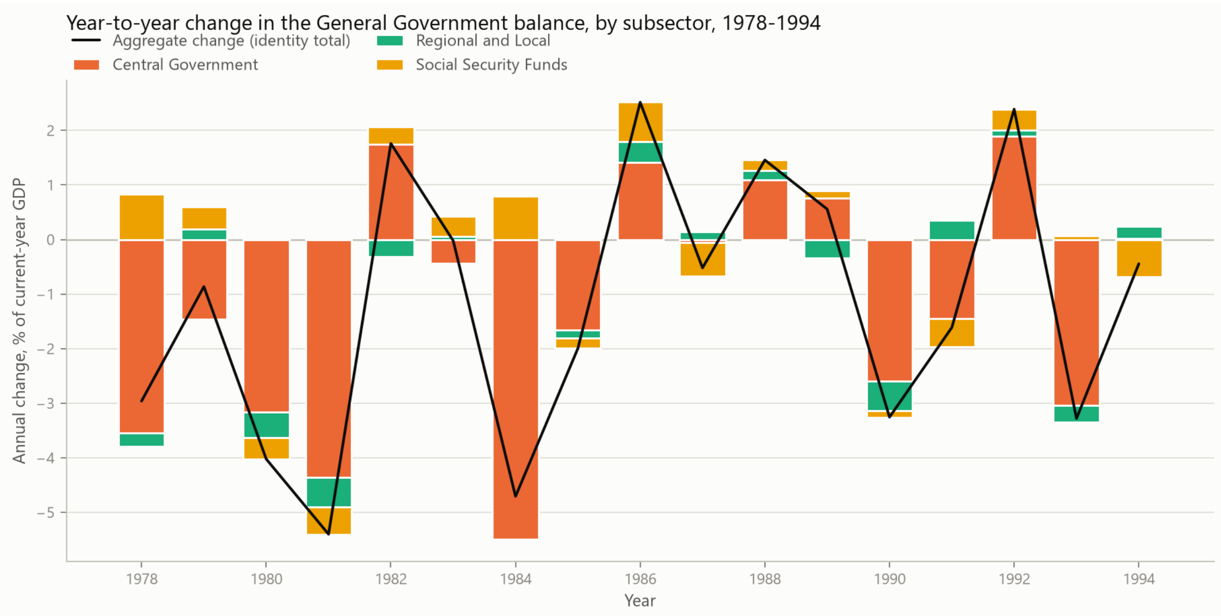

Figure 7: The same decomposition for 1978–1994, on the same scaling. The two windows are drawn as separate panels because a single panel spanning 1995 would place a vintage revision among the economic movements and give it equal visual weight.

Table 10: Recent General Government balance changes, decomposed into revenue and expenditure changes.

Year	Change in revenue (M EUR)	Change in expenditure (M EUR)	Change in balance (M EUR)
2018	4,794	-203	4,997
2019	3,513	2,391	1,122
2020	-4,176	7,636	-11,813
2021	9,188	3,742	5,447
2022	9,898	4,539	5,360
2023	10,394	6,607	3,787
2024	7,997	9,164	-1,167
2025	8,332	8,136	196

Table 11: The same episodes as in Table 9, with the subsector accounting for most of each move split into its own revenue and expenditure changes.

Year	Largest contributor	Its balance change (M EUR)	Δ revenue (M EUR)	Δ expenditure (M EUR)	Expenditure contribution (M EUR)
1981	Central Government	-385	395	780	-780
1984	Central Government	-922	616	1,538	-1,538
1980	Central Government	-233	388	621	-621
1993	Central Government	-2,057	-496	1,561	-1,561
1990	Central Government	-1,305	1,866	3,172	-3,172
2009	Central Government	-8,939	-3,467	5,473	-5,473
2020	Central Government	-10,224	-3,748	6,475	-6,475
2011	Central Government	5,937	1,209	-4,728	4,728
2015	Central Government	4,283	1,580	-2,703	2,703
2021	Central Government	5,570	6,855	1,285	-1,285

The split is of the named subsector, not of the aggregate, so the columns describe the same entity as the contributor column. Expenditure enters the balance negatively: the last column is minus the expenditure change, and it is that column which adds to the revenue change to give the subsector's balance change.

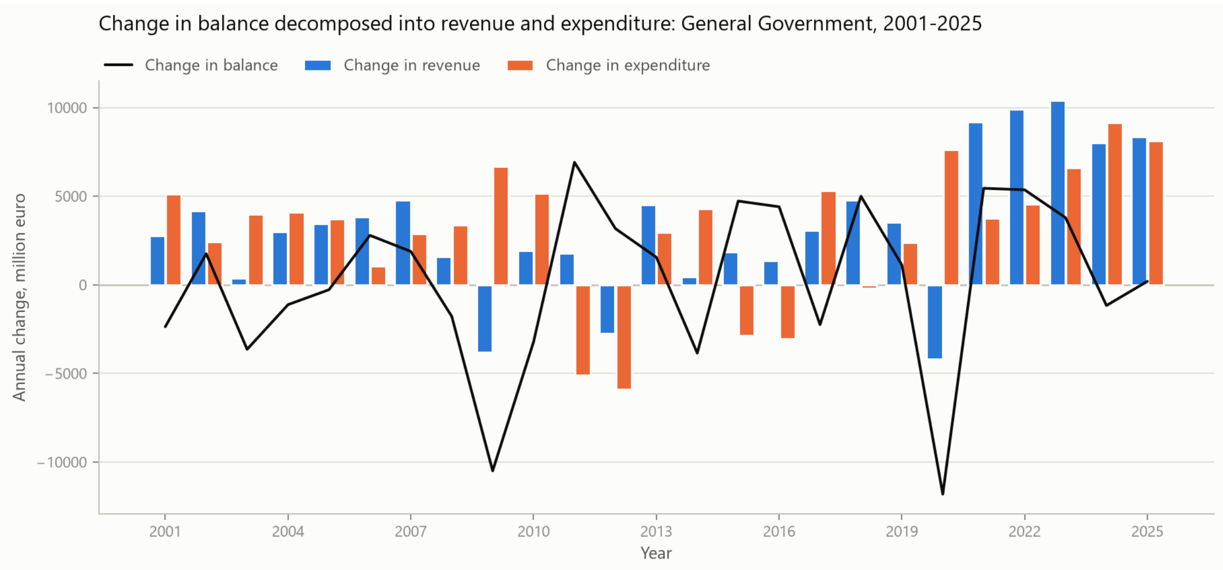

Figure 8: The identity drawn for General Government, 2001–2025: the annual change in the balance against the revenue and expenditure changes that compose it, in million euro.

Table 12: The three largest component movements behind each episode, for the subsector that dominates it, ranked by the absolute size of the contribution.

Regime	Year	Subsector	Side	Component	Change (M EUR)	Contribution to balance (M EUR)
1977–1994 historical	1981	Central Government	Expenditure	Current expenditure	702	-702
1977–1994 historical	1981	Central Government	Revenue	Current revenue	397	397
1977–1994 historical	1981	Central Government	Expenditure	Capital expenditure	78	-78
1977–1994 historical	1984	Central Government	Expenditure	Current expenditure	1,529	-1,529
1977–1994 historical	1984	Central Government	Revenue	Current revenue	591	591
1977–1994 historical	1984	Central Government	Revenue	Capital revenue	25	25
1977–1994 historical	1980	Central Government	Expenditure	Current expenditure	525	-525
1977–1994 historical	1980	Central Government	Revenue	Current revenue	380	380
1977–1994 historical	1980	Central Government	Expenditure	Capital expenditure	96	-96
1977–1994 historical	1993	Central Government	Expenditure	Current expenditure	1,494	-1,494
1977–1994 historical	1993	Central Government	Revenue	Current revenue	-486	-486
1977–1994 historical	1993	Central Government	Expenditure	Capital expenditure	68	-68
1977–1994 historical	1990	Central Government	Expenditure	Current expenditure	2,555	-2,555
1977–1994 historical	1990	Central Government	Revenue	Current revenue	1,663	1,663
1977–1994 historical	1990	Central Government	Expenditure	Capital expenditure	617	-617
1995–2025 modern	2009	Central Government	Revenue	Taxes	-4,248	-4,248
1995–2025 modern	2009	Central Government	Expenditure	Capital expenditure	1,571	-1,571
1995–2025 modern	2009	Central Government	Expenditure	Social transfers	1,346	-1,346
1995–2025 modern	2020	Central Government	Revenue	Taxes	-3,355	-3,355
1995–2025 modern	2020	Central Government	Expenditure	Other current expenditure	3,122	-3,122
1995–2025 modern	2020	Central Government	Expenditure	Capital expenditure	2,087	-2,087
1995–2025 modern	2011	Central Government	Expenditure	Capital expenditure	-4,594	4,594
1995–2025 modern	2011	Central Government	Expenditure	Interest	2,235	-2,235
1995–2025 modern	2011	Central Government	Revenue	Taxes	1,984	1,984
1995–2025 modern	2015	Central Government	Expenditure	Capital expenditure	-3,095	3,095
1995–2025 modern	2015	Central Government	Revenue	Taxes	1,775	1,775
1995–2025 modern	2015	Central Government	Expenditure	Social transfers	1,060	-1,060
1995–2025 modern	2021	Central Government	Revenue	Taxes	3,485	3,485
1995–2025 modern	2021	Central Government	Revenue	Sales and other current revenue	1,616	1,616
1995–2025 modern	2021	Central Government	Revenue	Capital revenue	1,460	1,460

The two source families resolve the accounts at different depths: the modern workbooks separate four revenue and seven expenditure components, the historical series only current from capital. Each sector-year uses the finer scheme it reports, so the modern period is never coarsened to match the historical one. Expenditure contributions are negated, so a falling capital expenditure appears as a positive contribution.

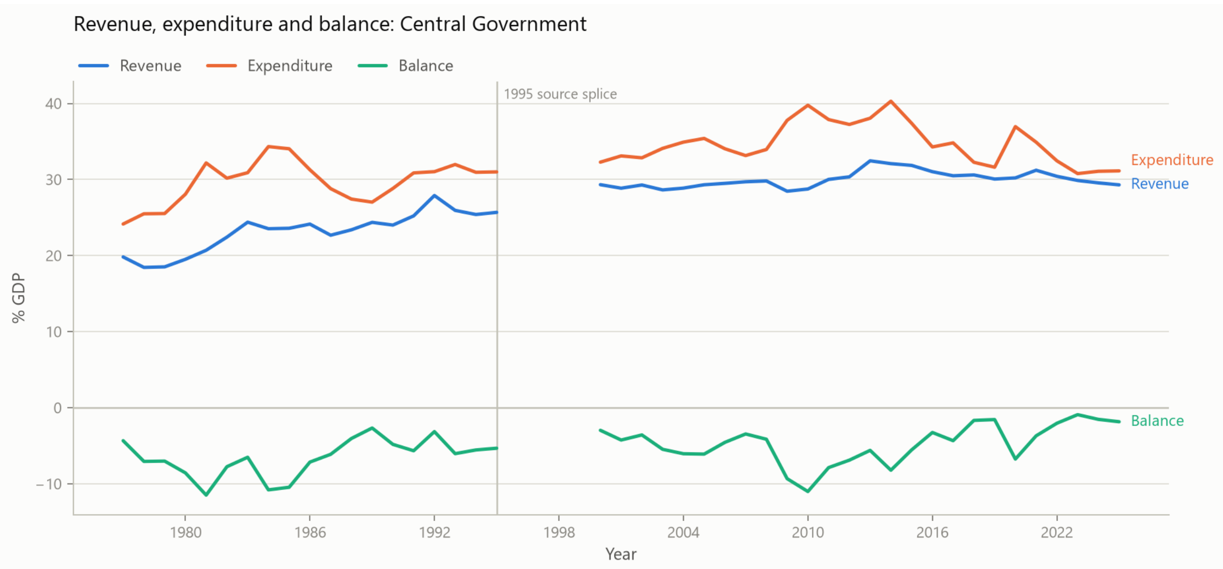

Figure 9: Central Government revenue, expenditure and balance as shares of GDP, 1977–2025. The line breaks at 1996–1999, where subsector components do not exist in either source; the gap is left open rather than interpolated. The vertical rule marks the 1995 splice.

The decomposition uses totals, so a change in revenue may reflect composition shifts that are visible only in the component columns of the account panel. It is not decomposed into policy and macroeconomic parts, which would require assumptions this report does not make.

6 Social Security Funds: Revenue Composition and Internal Systems

6.1 National-accounts view

The ESA 2010 Social Security Funds sector is the entity that appears in the B.9 identity. In 2025, social contributions were **67.07%** of total SSF revenue in the account table.

Table 13: Social Security Funds revenue composition, most recent years.

Year	Revenue (% GDP)	Contributions (% GDP)	Expenditure (% GDP)	Contribution share
2020	15.58	9.37	14.49	0.602
2021	14.92	9.36	13.78	0.628
2022	14.23	9.25	12.47	0.650
2023	14.01	9.38	11.89	0.670
2024	14.41	9.64	12.33	0.669
2025	14.77	9.91	12.47	0.671

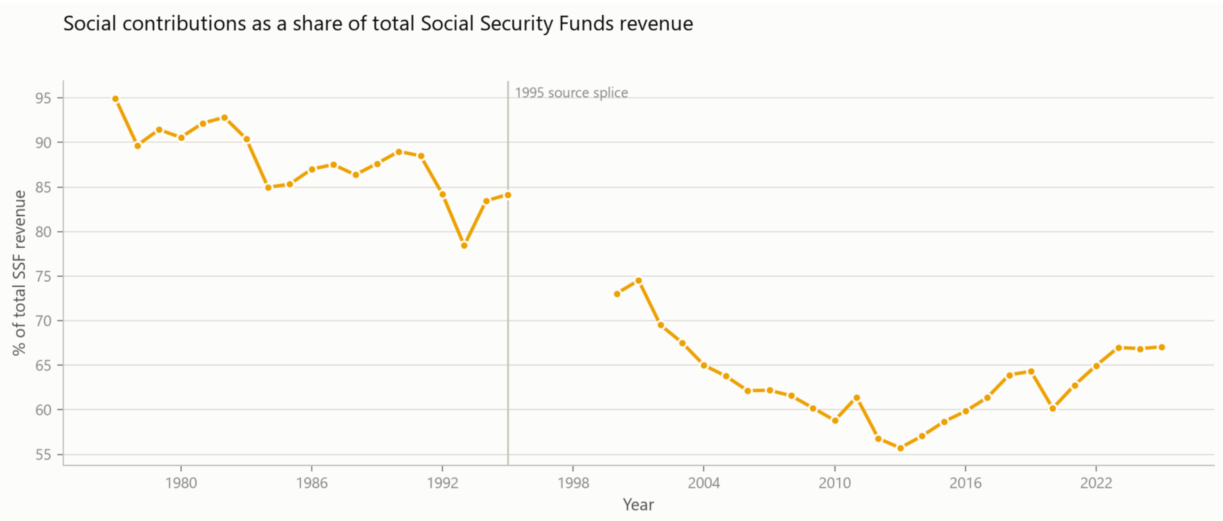

Figure 10: Social contributions as a share of total Social Security Funds revenue.

6.2 Where the annual change comes from

The balance identity differences exactly, so each annual change in the Social Security balance decomposes into the account movements that produced it. The revenue split is available for the whole detailed panel; the expenditure component detail exists only for the modern period, so social transfers are separated from the remainder there.

This locates a change in the accounts. It does not explain it: a contributory balance moves with employment, wages, contribution rates, entitlement rules and demographics at the same time, and separating those requires a model this report does not build.

Table 14: Contributions to the annual change in the Social Security balance. Each column carries the sign with which the term enters the balance, so the four add to the change.

Year	Change in balance (M EUR)	Social contributions (M EUR)	Other revenue (M EUR)	Social transfers (M EUR)	Other expenditure (M EUR)
2020	-778	311	2,197	-1,449	-1,837
2021	270	1,430	-454	-794	88
2022	1,831	2,289	138	-1,525	929
2023	1,421	2,806	337	-1,653	-69
2024	310	2,557	1,339	-3,017	-569
2025	1,034	2,468	1,087	-2,003	-518

Expenditure enters the balance negatively, so a rise in social transfers appears as a negative contribution. Reporting the raw expenditure change beside the balance change would invite adding two quantities of opposite sign. The identity closes to 0.0e+00 M EUR.

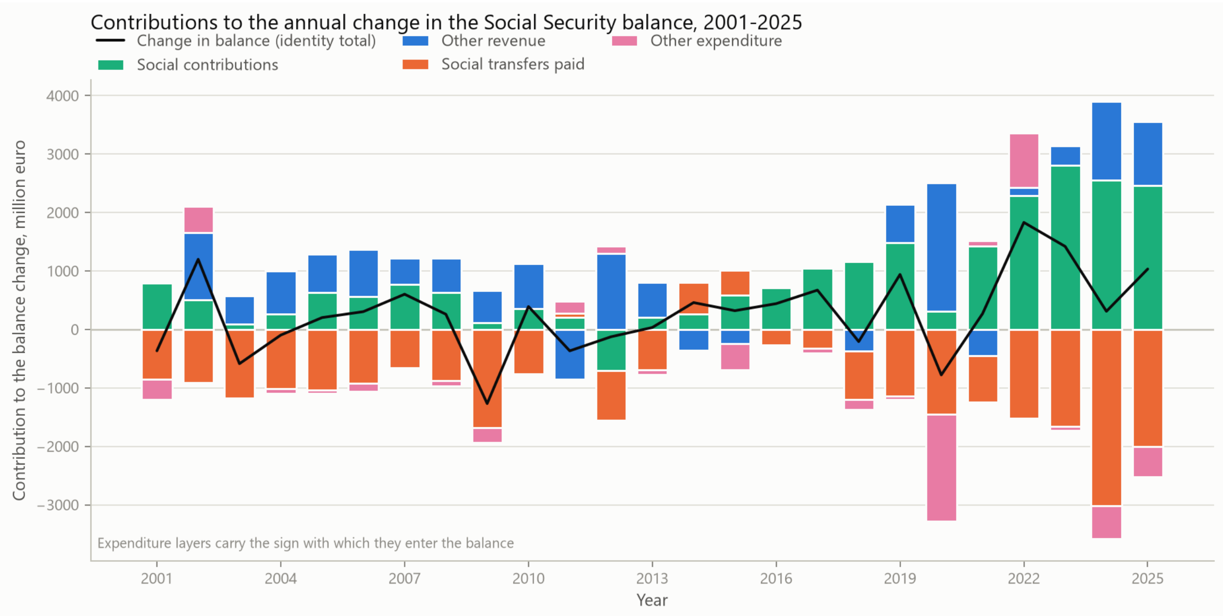

Figure 11: The same decomposition drawn, 2001 onward. Layers above zero raised the balance that year and layers below zero reduced it; the black line is the identity total.

6.3 A different boundary: the CFP budget systems

The CFP's Social Security report decomposes the system into the Previdential system, the Social Protection of Citizenship system, and special regimes. These are budget-execution aggregates with a different accounting boundary from the national accounts, so they are reported beside the ESA 2010 balance and never added to it. In 2025 the reported internal balances were **6,712 M EUR** (Previdential), **-55 M EUR** (Citizenship) and **0 M EUR** (special regimes).

Table 15: CFP Social Security budget systems: internal balances.

Year	Previdential (M EUR)	Citizenship (M EUR)	Special regimes (M EUR)
2019	2,333	443	0
2020	1,649	423	0
2021	1,695	653	0
2022	4,172	-110	0
2023	5,485	1	0
2024	5,840	-245	0
2025	6,712	-55	0

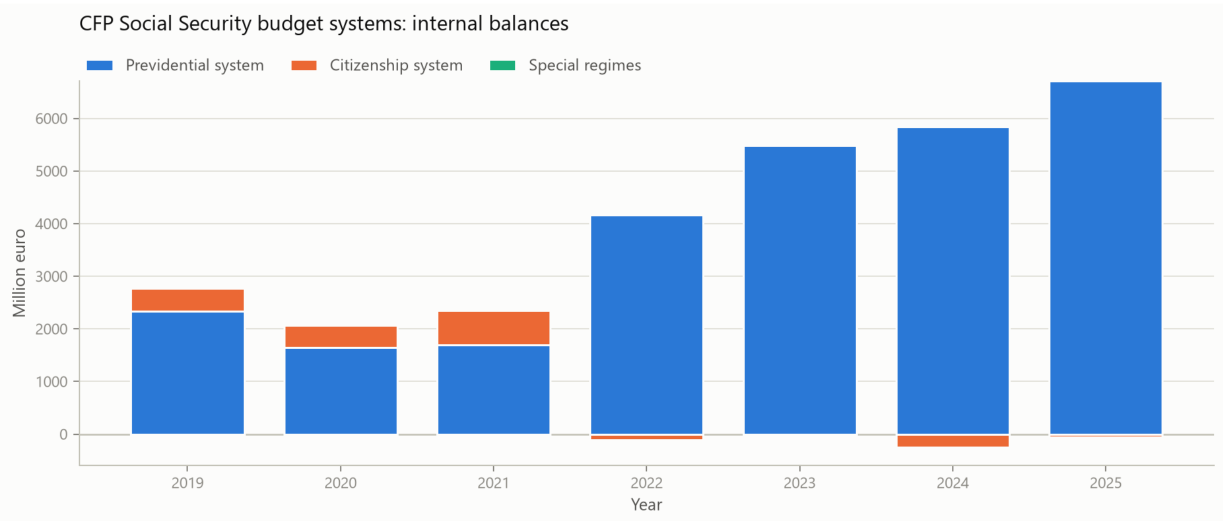

Figure 12: The CFP internal Social Security balances stacked by system, in million euro. The rise over the period is concentrated in the Previdential system, while the Citizenship system moves between small positive and small negative balances.

Across the years the CFP publishes, the Previdential balance moves from **2,333 M EUR** in 2019 to **6,712 M EUR** in 2025, while the Citizenship balance stays small in both directions. The movement in the internal balances is therefore concentrated in one of the two systems. This is a description of the published series and not an account of what caused it.

For the detailed 2025 budget table, previdential contributions account for **92.27%** of previdential revenue. State transfers are not subtracted from the Social Security balance to produce an underlying balance: which transfer finances which statutory responsibility is a legal question, not an accounting one.

6.4 Why the two boundaries must not be interchanged

The two Social Security balances are close but never equal. In 2025 the ESA 2010 balance was **7,065 M EUR** while the three budget systems summed to **6,657 M EUR**, a difference of **408 M EUR**. A difference of the same order is present in every year the two overlap.

Table 16: The two accounting boundaries side by side. The columns are not added, netted or reconciled: they are different objects, and the difference column measures how far apart they are.

Year	ESA 2010 balance (M EUR)	Budget-system total (M EUR)	Difference (M EUR)	Difference as share of ESA balance
2019	2,975	2,776	199	0.067
2020	2,198	2,072	126	0.057
2021	2,468	2,348	120	0.049
2022	4,299	4,062	237	0.055
2023	5,720	5,486	234	0.041
2024	6,031	5,595	436	0.072
2025	7,065	6,657	408	0.058

The difference is non-zero in every overlapping year. That is the reason a Social Security figure quoted from the budget documents cannot be substituted for the one that enters the national-accounts identity.

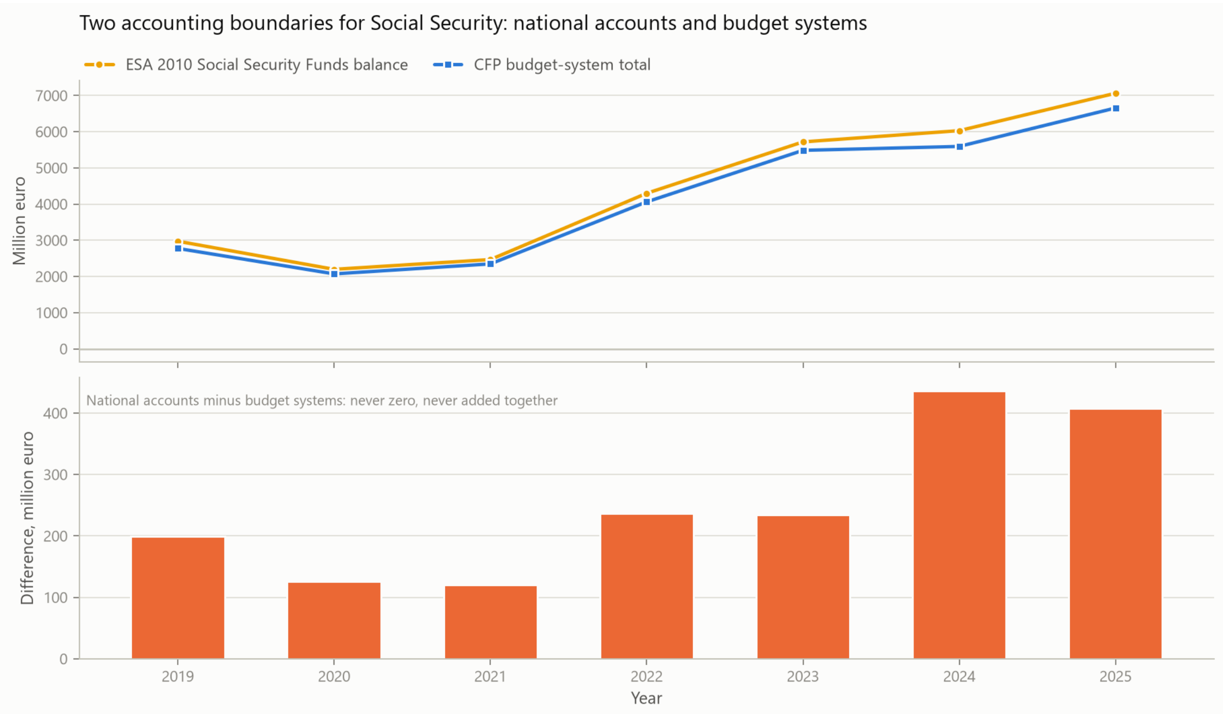

Figure 13: The same comparison drawn. The upper panel carries both balances on a shared axis, where they are nearly indistinguishable; the lower panel carries the difference on its own axis, which is invisible at the scale of the levels.

7 Contributions and the Aggregate Employee Wage Bill

The Social Security results so far locate a movement inside the fiscal accounts. They do not relate it to anything outside them. Contributions are levied predominantly on wages, so the aggregate employee wage bill is their natural reference base, $W_t = N_t \bar{w}_t$, with N employees and $\bar{w}$ the average wage. It is a reference base and not the integral base of the levy: some recorded contributions fall on other bases and some are imputed, so W does not exhaust what C is charged on. Writing the contributions-to-wage-bill ratio as $\tau_t = C_t/W_t$, the change in contributions decomposes exactly:

$$\Delta C_t = \tau_{t-1}\Delta W_t + W_{t-1}\Delta\tau_t + \Delta W_t \Delta\tau_t.$$

The wage bill and employment come from the Portuguese national accounts compiled by INE. τ is *not* a statutory contribution rate: its numerator contains revenue raised on bases its denominator does not measure, so the ratio moves with coverage, compliance and composition as well as with legislated rates. It stands at **0.265** in 2025 against **0.260** in 1995.

Table 17: The annual change in Social Security contributions, split into the movement of the wage bill, the movement of the contributions-to-wage-bill ratio, and their interaction.

Year	Change in contributions (M EUR)	From the wage bill (M EUR)	From the ratio (M EUR)	Interaction (M EUR)
2018	1,166	947	206	12
2019	1,484	1,014	443	26
2020	311	28	283	0
2021	1,430	1,268	152	10
2022	2,289	2,110	163	17
2023	2,806	2,685	107	13
2024	2,557	2,056	463	38
2025	2,468	1,870	560	38

The interaction term is carried rather than dropped or shared between the two factors, because either would make the decomposition inexact. It closes to 3.2e-12 M EUR.

In 2025 contributions rose by **2,468 M EUR**, of which **1,870 M EUR** came from the wage bill and **560 M EUR** from the ratio, with the remainder their interaction. A second and separate application of the same identity splits the change in the wage bill itself. The two have different totals, and the employment and average-wage terms below are components of ΔW rather than sub-components of the wage-bill term above.

Table 18: And what moved the wage bill itself: the number of employees against the average wage per employee. This decomposes the change in the wage bill, not the wage-bill component of the change in contributions; the two have different totals.

Year	Change in the wage bill (M EUR)	From employment (M EUR)	From average wages (M EUR)	Interaction (M EUR)
2018	3,998	1,423	2,521	54
2019	4,225	478	3,722	25
2020	116	-1,400	1,544	-29
2021	5,071	852	4,172	47
2022	8,372	2,985	5,194	193
2023	10,572	2,000	8,383	189
2024	8,058	992	6,996	70
2025	7,196	2,253	4,842	102

The interaction terms are carried rather than distributed. An equally standard convention evaluates each factor at the midpoint of the two years, which splits the interaction evenly between them and is exact in two terms rather than three. It is reported alongside so the reading does not rest on the convention.

Table 19: The same change under the symmetric convention, which evaluates each factor at the midpoint of the two years and thereby splits the interaction evenly between them.

Year	Change in contributions (M EUR)	From the wage bill (M EUR)	From the ratio (M EUR)	Wage-bill share
2018	1,166	953	213	0.82
2019	1,484	1,027	456	0.69
2020	311	29	283	0.09
2021	1,430	1,273	157	0.89
2022	2,289	2,118	171	0.93
2023	2,806	2,692	114	0.96
2024	2,557	2,075	482	0.81
2025	2,468	1,889	579	0.77

Exact in two terms rather than three. Neither convention is more correct; the three-term form is reported above because carrying the interaction rather than distributing it is the more conservative presentation.

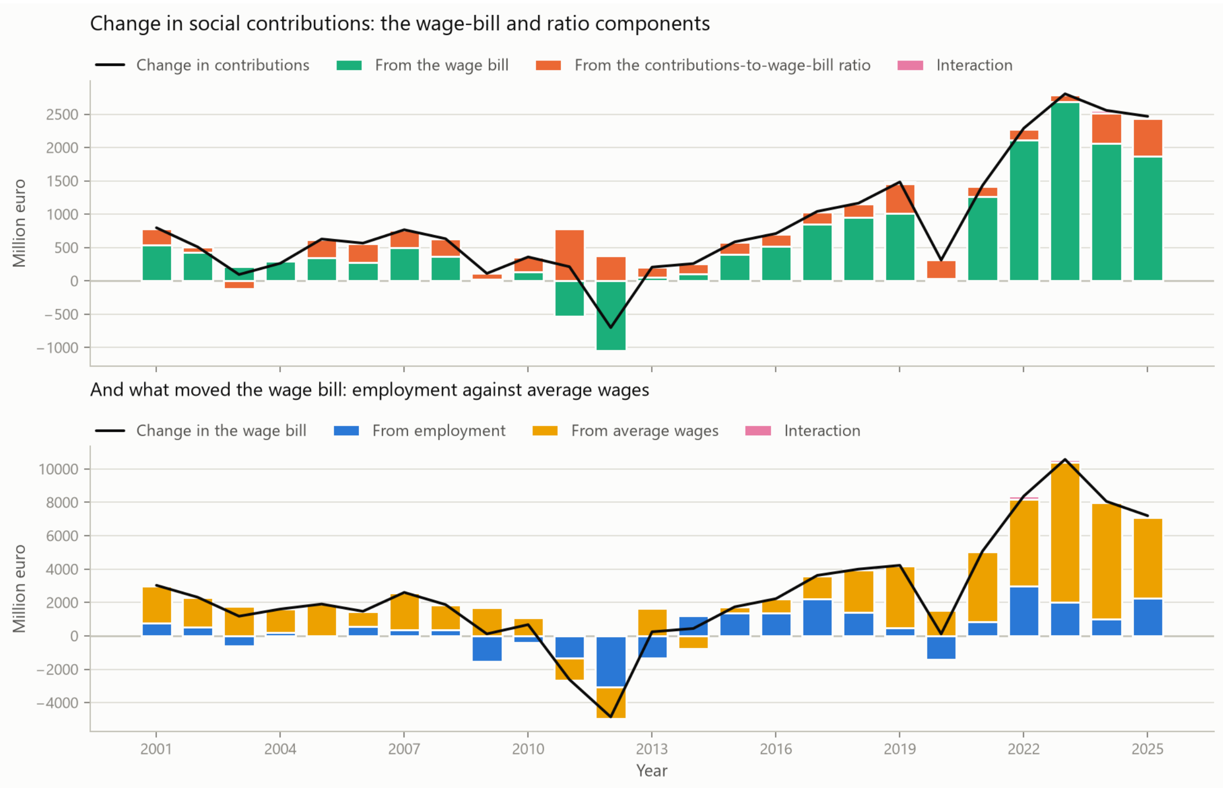

Figure 14: Contributions and their reference base. Upper panel: the change in contributions split into a wage-bill component, a ratio component and their interaction. Lower panel: the change in the wage bill split into an employment component, an average-wage component and their interaction.

7.1 A companion regression

Regressing the annual change in contributions on the annual change in the wage bill over 25 adjacent-year pairs gives a slope of **0.248** with a HAC standard error of 0.022 and an R^2 of **0.926**. The slope sits close to the mean contributions-to-wage-bill ratio of 0.221, which is consistent with a stable ratio over the sample but is not implied by the identity, and the fit is far tighter than the nominal-GDP specification in Appendix D: the regressor here is a quantity the levy is charged on rather than an aggregate that merely correlates with it. It remains a descriptive statistic and no identification is claimed.

The decomposition is exact and descriptive. It does not establish that the wage bill produced the movement in contributions, and it identifies nothing about why employment or wages moved.

8 Primary Balance and Interest

The primary balance is reconstructed as $PB_{i,t} = B_{i,t} + I_{i,t}$, where I is interest expenditure, and then checked against the published primary-balance row. The identity holds to **0.0e+00 M EUR**, so the separation of interest from the remaining balance is arithmetic rather than estimated.

For Central Government in 2025 the headline balance was **-5,636 M EUR**, interest expenditure was **6,364 M EUR**, and the recomputed primary balance was **728 M EUR**.

Table 20: Central Government headline balance, interest and primary balance.

Year	Balance (M EUR)	Interest (M EUR)	Primary balance (M EUR)	Interest (% GDP)	Primary balance (% GDP)
2018	-3,407	7,101	3,694	3.46	1.80
2019	-3,331	6,516	3,185	3.04	1.48
2020	-13,555	5,963	-7,592	2.97	-3.78
2021	-7,985	5,358	-2,627	2.47	-1.21
2022	-4,966	4,878	-88	2.00	-0.04
2023	-2,447	5,797	3,349	2.14	1.24
2024	-4,424	6,281	1,857	2.17	0.64
2025	-5,636	6,364	728	2.07	0.24

8.1 The headline sign is not the primary sign

Central Government records a negative B.9 in every year of the canonical panel. Read alone, that invites the conclusion that the subsector is in deficit on every measure. It is not. Over the **45 years** for which interest expenditure is available, the Central Government headline balance is negative in **45** of them, while the primary balance is positive in **15**: 1986, 1987, 1988, 1989, 1990, 1991, 1992, 1994, 1995, 2016, 2018, 2019, 2023, 2024 and 2025.

The two statements are both descriptive and they are not in conflict. The headline balance is negative throughout; the primary balance, which excludes interest by construction, is positive in a non-trivial minority of the observed years. Interest peaked at **7.73% of GDP** and averaged **4.17%** across the panel, which is the arithmetic that separates them.

Those statistics pool the two source families. The sign counts survive that, because a classification is not a magnitude, but the interest means do not: interest is a magnitude and the two families differ materially in it. Both are therefore reported by family as well, with the years each family spans in that sector printed alongside, since the three subsectors carry detailed accounts for 1977–1995 and 2000–2025 while General Government carries them continuously.

Interest is overwhelmingly a Central Government item, which is why the primary and headline balances of the other subsectors nearly coincide. The primary balance excludes interest by construction: it is not a measure of discretionary policy and not a cyclically adjusted balance, and it reflects the debt stock and financing conditions inherited from earlier periods. A positive

Table 21: Headline against primary balance sign frequencies, over the sector-years for which interest expenditure is available.

Sector	N	Headline < 0	Primary > 0	Mean interest (% GDP)	Peak interest (% GDP)
General Government	49	45	22	4.16	7.89
Central Government	45	45	15	4.17	7.73
Regional and Local	45	29	20	0.13	0.62
Social Security Funds	45	6	39	0.00	0.04

The panel here is the detailed account panel, so the three subsectors have 45 observations rather than the 49 of the canonical balance panel; the 1996–1999 components are missing.

Table 22: The same frequencies inside each source family. The counts may be pooled; the interest magnitudes may not.

Source family	Sector	First year	Last year	N	Headline < 0	Primary > 0	Mean interest (% GDP)	Peak interest (% GDP)
Historical long series	General Government	1977	1994	18	18	9	5.66	7.89
ESA 2010 modern	General Government	1995	2025	31	27	13	3.29	5.54
Historical long series	Central Government	1977	1995	19	19	9	5.52	7.73
ESA 2010 modern	Central Government	2000	2025	26	26	6	3.19	4.88
Historical long series	Regional and Local	1977	1995	19	13	10	0.18	0.62
ESA 2010 modern	Regional and Local	2000	2025	26	16	10	0.09	0.18
Historical long series	Social Security Funds	1977	1995	19	5	14	0.01	0.04
ESA 2010 modern	Social Security Funds	2000	2025	26	1	25	0.00	0.00

The first and last year are printed because a source family does not span the same years in every sector: General Government carries detailed accounts continuously, while the three subsectors carry them for 1977–1995 and 2000–2025 only. Interest is a magnitude and the families differ materially in it, so it must not be pooled; the sign counts are reported both ways because the classification survives the splice where the magnitudes do not.

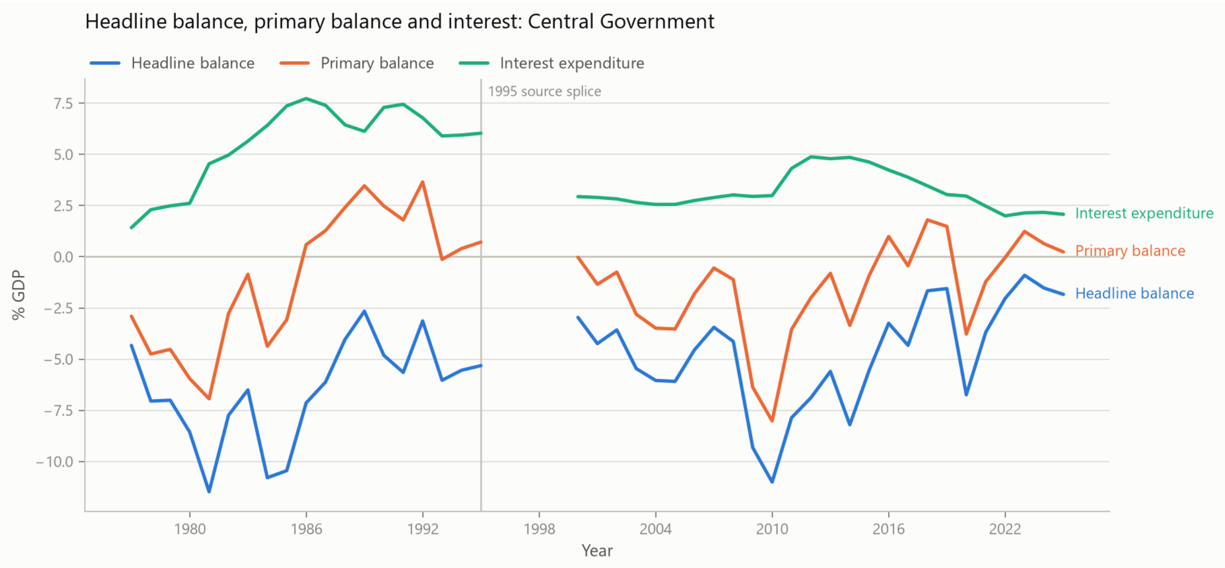

Figure 15: Central Government headline balance, primary balance and interest expenditure as shares of GDP, over every year for which interest is available. The primary balance crosses zero repeatedly while the headline balance does not; the line breaks at 1996–1999 where subsector components are missing.

primary balance is therefore not evidence of a sustainable position, and a negative headline balance is not evidence of an unsustainable one.

9 Persistence and Structural Mean Shifts

9.1 Sign frequencies and runs

Table 23: Sign frequency, average magnitude and longest runs by subsector, pooled over the whole panel. The pooled means are reported for completeness only; see Table 25 for the regime split.

Sector	N	Positive	Negative	Mean (% GDP)	Median (% GDP)	Longest + run	Longest - run
General Government	49	4	45	-4.915	-5.028	3	42
Central Government	49	0	49	-5.381	-5.396	0	49
Regional and Local	49	18	31	-0.157	-0.124	8	12
Social Security Funds	49	43	6	0.623	0.495	24	2

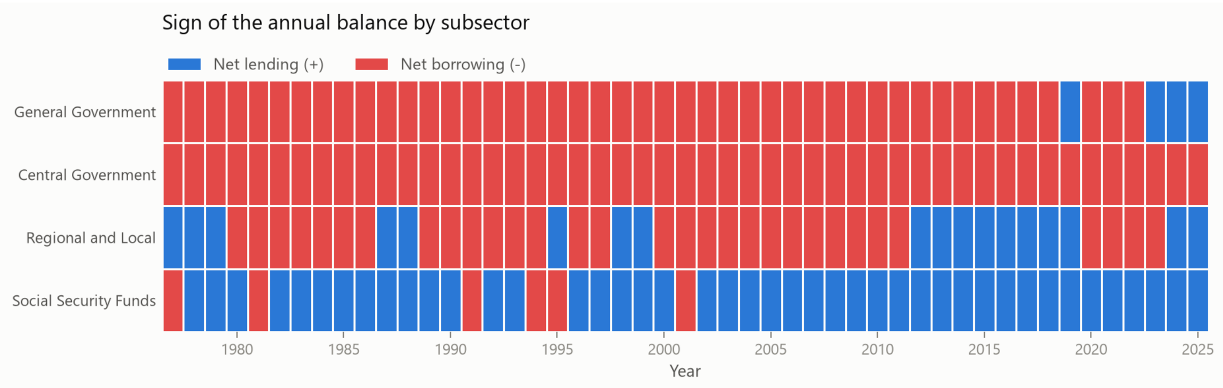

Figure 16: Sign of the annual balance by subsector and year.

Table 24: Empirical one-year sign transition frequencies. Rows sum to one.

Sector	From	To negative	To positive
General Government	negative	0.956	0.044
General Government	positive	0.333	0.667
Central Government	negative	1.000	0.000
Regional and Local	negative	0.839	0.161
Regional and Local	positive	0.294	0.706
Social Security Funds	negative	0.167	0.833
Social Security Funds	positive	0.095	0.905

These are empirical frequencies over the transitions actually observed, not a fitted Markov model: no standard errors and no stationarity test are claimed, and a state that never occurs in the sample has no estimated row. Runs that span 1995 also span the statistical splice.

9.2 Pooled averages describe neither regime

The magnitudes in Table 23 average across the 1995 splice, and the two regimes differ enough that the pooled figure is not a good description of either. The aggregate balance averages **-6.43% of GDP** over 1977–1994 and **-4.04%** over 1995–2025, against a pooled **-4.91%** that falls between them and corresponds to no observed period. The regime split is therefore the form in which magnitudes should be read.

Sign counts are far more robust to pooling, because a sign does not depend on the level convention of the vintage. Both are reported per regime below so they are read on one basis.

Table 25: The same sign counts and magnitudes computed inside each statistical regime rather than pooled across both.

Regime	Sector	N	Positive	Negative	Mean (% GDP)	Median (% GDP)
1977–1994 historical	General Government	18	0	18	-6.429	-6.033
1995–2025 modern	General Government	31	4	27	-4.036	-4.167
1977–1994 historical	Central Government	18	0	18	-6.614	-6.312
1995–2025 modern	Central Government	31	0	31	-4.666	-4.327
1977–1994 historical	Regional and Local	18	5	13	-0.275	-0.190
1995–2025 modern	Regional and Local	31	13	18	-0.088	-0.103
1977–1994 historical	Social Security Funds	18	14	4	0.460	0.474
1995–2025 modern	Social Security Funds	31	29	2	0.718	0.610

Runs are not recomputed per regime: a run is a property of the uninterrupted series, and truncating it at a window boundary would report the length of the window.

9.3 Structural mean shifts

Fewer than fifty annual observations, split across two statistical regimes, do not support a flexible change-point model. The specification is therefore deliberately modest: a piecewise-constant mean with at most two breaks per regime, a minimum segment length of five years, exact dynamic-programming minimisation of within-segment squared error, and BIC selection, which may select zero breaks. Detection runs separately on 1977–1994 and 1995–2025, so the known splice is not a candidate economic break by construction.

Table 26: Preferred piecewise-constant mean specification, by statistical regime. The BIC margin is how much better the selected break count scores than the next-best count.

Regime	Sector	N	Breaks	Break years	Segment means (% GDP)	BIC margin
1977–1994 historical	General Government	18	1	1986	-8.0942, -4.7630	3.43
1995–2025 modern	General Government	31	2	2009, 2015	-4.3662, -7.9664, -1.4712	5.17
1977–1994 historical	Central Government	18	1	1987	-8.1056, -4.7492	5.21
1995–2025 modern	Central Government	31	2	2009, 2016	-4.4779, -7.7736, -2.7526	10.94
1977–1994 historical	Regional and Local	18	0	–	-0.2750	4.26
1995–2025 modern	Regional and Local	31	2	2000, 2011	-0.0006, -0.4598, 0.1550	3.62
1977–1994 historical	Social Security Funds	18	0	–	0.4602	1.15
1995–2025 modern	Social Security Funds	31	2	2016, 2021	0.3522, 1.0925, 1.8804	3.19

9.4 How firm are those dates?

With eighteen or thirty-one annual observations per regime, a single selected date should not be read as determined. Two guards are reported. The BIC margin in Table 26 shows how decisively the selected break count beat the alternatives, and the grid in Table 27 re-runs detection across all twelve combinations of the two tuning parameters.

In 6 of the 8 series the preferred specification returns the modal set of dates. In the remaining 2 it does not: Regional and Local in the 1995–2025 modern regime, Social Security Funds in the 1995–2025 modern regime. For those series the selected dates follow the tuning choice rather than the series, and the modal column in Table 27 is the more reliable summary.

The least stable series is **Regional and Local** in the 1995–2025 modern regime, where the modal dates hold in only **33%** of the grid across **6 distinct date sets**.

Table 27: Sensitivity of the detected breaks to the two tuning choices, over a grid of twelve specifications per series: minimum segment length in 4, 5, 6, 7 crossed with a maximum of 1, 2 or 3 breaks.

Regime	Sector	Specifications	Modal breaks	Share at modal count	Modal dates	Share at modal dates	Distinct date sets
1977–1994 historical	General Government	12	1	0.83	1986	0.83	2
1995–2025 modern	General Government	12	2	0.58	2009, 2015	0.42	4
1977–1994 historical	Central Government	12	1	1.00	1987	1.00	1
1995–2025 modern	Central Government	12	2	0.67	2009, 2016	0.67	2
1977–1994 historical	Regional and Local	12	0	0.83	–	0.83	2
1995–2025 modern	Regional and Local	12	2	0.42	2012	0.33	6
1977–1994 historical	Social Security Funds	12	0	1.00	–	1.00	1
1995–2025 modern	Social Security Funds	12	1	0.67	2017	0.67	3

Neither tuning parameter is estimated from the data. A date that survives only one of their values is a property of that choice, not of the series, and the share columns are what distinguishes the two cases.

The dates are accordingly stated as candidates rather than as findings: the preferred specification identifies shifts around the years listed, and the share columns say how much of the specification grid agrees. A break year should not be quoted from this report without its share. This report attaches no historical cause to any of them, and a minimum segment length of five years means shifts near the end of the sample cannot be detected yet.

10 European Benchmark

Every other section of this report describes one country, which cannot establish whether what it describes is unusual. ESA 2010 requires the same subsector breakdown from every reporter, so the comparison is available. The definitions used here are the domestic ones, with three adjustments.

1. The non-Social-Security aggregate is central *plus state* plus local government. Portugal has no state tier; several reporters do, and omitting it would leave their identity open.
2. Ratios are computed in national currency. The published shares of GDP carry one decimal, which is too coarse a denominator for a ratio.
3. A reporter enters only with at least fifteen complete years, so a frequency over a handful of years is not compared with one over thirty.

This leaves **28 reporters** over 1995–2025. Eurostat's Portuguese rows agree with the domestic panel to rounding, so this source doubles as an independent check on the extraction.

The answer differs across the findings. A Central Government deficit is recorded in **100%** of Portuguese years, but **9 of 28 reporters** record one in every year they cover: a persistently deficit-running central tier is a common European pattern. The Social Security surplus is different, occurring in **93.5%** of Portuguese years against a cross-country median of **68.9%**, and averaging **0.71% of GDP** against a median of **0.17%**.

The sharpest comparison is the composition of a surplus year. Across the **22 reporters** with at least one aggregate surplus, a negative non-Social-Security balance accompanies the surplus in **54 of 198** surplus country-years, or **27%**. It is a minority pattern. **1 of those reporters** show it in every one of their surplus years, Portugal among them, on **4 surplus years** — a count small enough that it is stated beside the claim.

This is a distribution, not a test. Reporters differ in whether they operate a state tier, in how contributory schemes are assigned between tiers, in pension-system maturity and in how transfers are routed, and none of that is held constant here.

11 Methodological Limitations

1. **1995 is a source and methodology splice**, not an economic event. Both vintages are retained in `data/interim/methodology_overlap_1995.csv` and nothing is smoothed

Table 28: Subsector composition across European reporters, ordered by the frequency of a Social Security surplus.

Reporter	N	Central < 0	SSF > 0	Mean SSF (% GDP)	Surplus years	of which non-SSF < 0	Median offset
FI	31	0.774	1.000	2.45	11	6	0.540
LU	31	0.645	1.000	1.65	24	10	0.892
DK	31	0.355	0.935	0.05	21	0	0.008
CY	31	0.806	0.935	2.23	9	4	0.446
PT	31	1.000	0.935	0.71	4	4	0.141
IT	31	1.000	0.839	0.18	0	0	0.064
EL	31	0.903	0.839	0.97	6	3	0.206
EE	31	0.613	0.839	0.25	12	2	0.306
IS	28	0.536	0.821	0.16	10	0	0.043
SE	31	0.581	0.806	0.19	14	2	0.225
SK	31	1.000	0.742	0.42	0	0	0.061
BG	31	0.613	0.710	0.17	13	2	0.095
RO	31	1.000	0.710	0.13	0	0	0.084
CH	30	0.533	0.700	0.16	17	5	0.058
LV	31	0.968	0.677	0.35	2	0	0.316
LT	31	1.000	0.613	0.25	4	3	0.419
DE	31	0.774	0.613	0.04	8	2	0.085
NL	31	0.806	0.581	0.17	8	4	0.310
IE	31	0.516	0.581	0.09	16	3	0.167
BE	31	1.000	0.581	0.06	3	2	0.072
AT	31	0.968	0.548	0.02	2	0	0.056
HU	31	1.000	0.548	0.02	0	0	0.045
SI	31	0.935	0.516	-0.05	3	1	0.043
PL	31	1.000	0.484	0.24	0	0	0.263
HR	31	0.871	0.452	-0.12	4	1	0.124
FR	31	1.000	0.419	-0.20	0	0	0.072
CZ	31	0.935	0.387	-0.04	4	0	0.051
ES	31	0.903	0.355	-0.27	3	0	0.464

The non-Social-Security aggregate is central plus state plus local government, so federal reporters are treated consistently with unitary ones. Reporters with fewer than fifteen complete years are excluded.

Table 29: Portugal's position in each cross-country distribution.

Metric	Portugal	Cross-country median	Minimum	Maximum	Percentile
Share of years with a Central Government deficit	1.000	0.903	0.355	1.000	68
Share of years with a Social Security surplus	0.935	0.689	0.355	1.000	82
Share of years with an aggregate surplus	0.129	0.129	0.000	0.774	39
Mean Social Security balance (% GDP)	0.706	0.168	-0.271	2.448	82
Median offset ratio, where defined	0.141	0.110	0.008	0.892	54

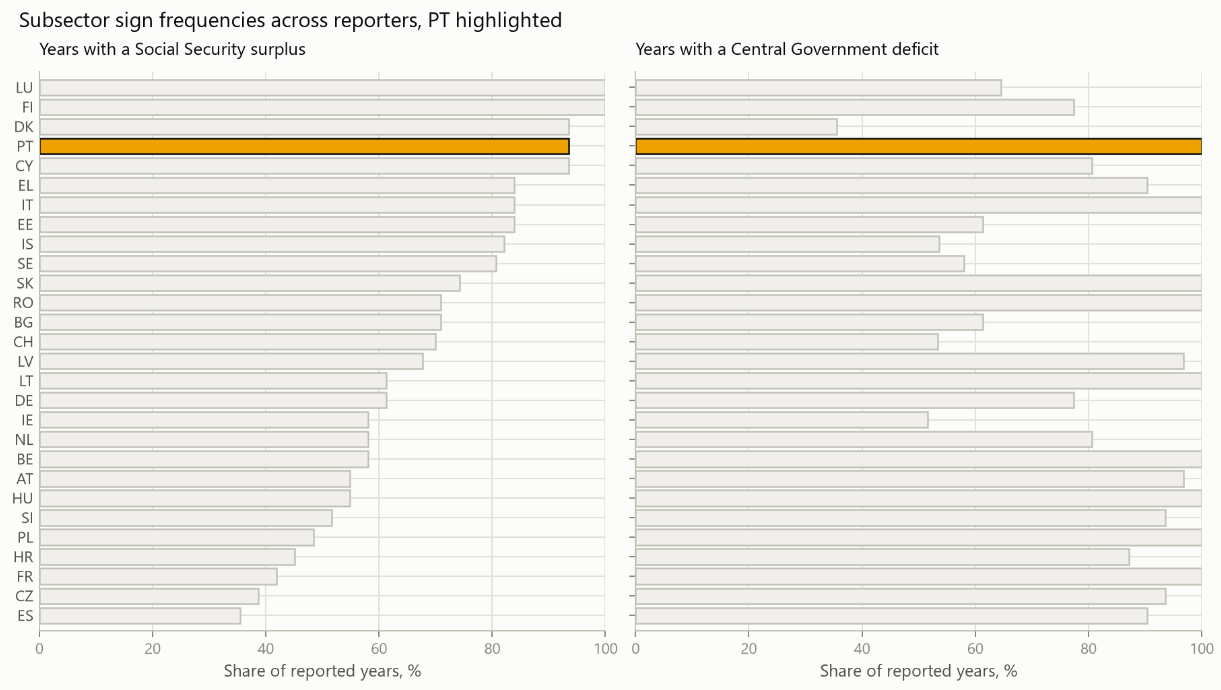

Figure 17: Subsector sign frequencies across reporters, Portugal highlighted and reporters ordered by the frequency of a Social Security surplus.

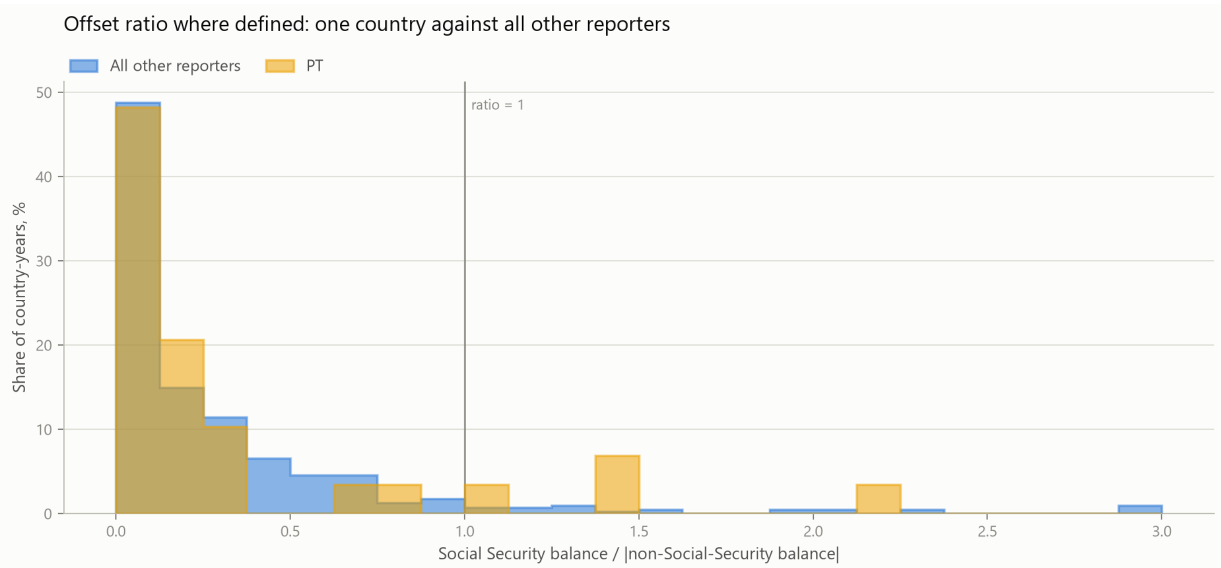

Figure 18: Offset ratio where defined: Portugal against all other reporters pooled, as a share of each group's defined country-years. Values above three are clipped.

across the boundary. No level statistic is averaged across it, and no annual change is computed through it.

2. **Detailed subsector accounts have no 1996–1999 observations.** No interpolation is performed, no statistic is computed across the gap, and every figure drawn from the account panel breaks its line there. The canonical B.9 panel is complete and is unaffected.
3. **The most recent years are provisional.** They carry the publisher's flag through the canonical panel and will be revised by a later vintage. They are also the years this report discusses in most detail.
4. **Identity closure is not source agreement.** The identities close to rounding while two published sources still disagree by up to about 67 million euro on the same subsector-year. Both tests are reported, and neither substitutes for the other.
5. **Nominal GDP growth is not an output gap**, and the co-movement appendix must not be read as a cyclically adjusted balance. Its dependent variable and regressor share a construction, which is one of the reasons it is confined to an appendix.
6. **Social Security national accounts and budget-system accounts have different boundaries.** They are never merged, netted or reconciled into a single series, and the difference between them is non-zero in every overlapping year.
7. **The fixed-investment diagnostic is not an official balance concept.**
8. **Structural-break dates are candidates, not findings.** They are reported with a BIC margin and a sensitivity grid precisely because a single selected date from eighteen or thirty-one observations is not determined.
9. **A positive primary balance is not a sustainability result.** It excludes interest by construction and says nothing about the debt path.
10. **Accounting contribution is not causation.** An arithmetic contribution to an aggregate balance is not evidence about intent or responsibility.
11. **Identity and reconciliation residuals validate the extraction**, not the underlying official statistics.
12. All interpretation is restricted to accounting, statistical and economic relationships directly supported by the bundled data.

12 Reproducibility

The repository preserves the raw source workbooks, source-specific intermediate CSVs, the canonical processed panels, every calculated analysis table, the figures, the executed notebooks and a SHA-256 digest of each bundled raw file. The default rebuild is

```
poetry install
make all
make pdf
```

which runs the deterministic pipeline, executes the notebooks with their outputs stored in place, runs the test suite and rebuilds the committed PDF. This document is regenerated from the persisted outputs rather than from hard-coded conclusions, and it is built from repository version **0.2.0**.

A Fixed-Capital-Formation Diagnostic

This appendix reports a quantity that is not an official balance concept. It is retained because the pipeline computes it and because the question it answers is a reasonable one to ask; it is here rather than in the body because no result above depends on it, and because a non-official indicator sitting among B.9 results invites being read as one.

The diagnostic is

$$B_{i,t}^{before\ GFCF} = B_{i,t} + GFCF_{i,t},$$

which answers one narrow question: how large is public investment relative to the recorded balance? It is not a golden-rule balance, not a structural balance, and not any published indicator; no fiscal rule in this report is evaluated against it.

Table 30: Fixed-capital-formation diagnostic by sector, 2025.

Sector	Balance (M EUR)	GFCF (M EUR)	Balance before GFCF (M EUR)	GFCF (% GDP)
General Government	2,059	9,167	11,226	2.99
Central Government	-5,636	4,828	-808	1.57
Regional and Local	630	4,236	4,866	1.38
Social Security Funds	7,065	102	7,167	0.03

Two caveats attach to the diagnostic. GFCF is gross, so no consumption of fixed capital is netted off and the measure overstates the change in the public capital stock. And investment is lumpy: a single-year ratio can be dominated by one project or one reclassification.

B Debt and Stock-Flow Adjustment

Debt dynamics are a different subject from balance composition, which is why this is an appendix. It belongs in the document because it establishes a limit on everything in the body: a result about the composition of a balance is not a result about debt.

The modern CFP tables support the reconciliation

$$\Delta Debt_t = -B_t + SFA_t,$$

where the stock-flow adjustment absorbs everything that changes debt without passing through the balance: financial-asset transactions, valuation effects, timing differences and statistical adjustments. The reconciliation closes to **0.0e+00 M EUR**.

In 2025, General Government Maastricht debt was **89.67% of GDP** and the stock-flow adjustment was **2.03% of GDP**. In several years the adjustment is the larger of the two terms, which is the reason for running the reconciliation at all.

Table 31: General Government debt and stock-flow adjustment, most recent years.

Year	Balance (M EUR)	Change in debt (M EUR)	Stock-flow adj. (M EUR)	Debt (% GDP)	Stock-flow adj. (% GDP)
2016	-3,622	9,449	5,827	131.18	3.13
2017	-5,870	1,904	-3,966	126.03	-2.03
2018	-873	1,879	1,006	121.11	0.49
2019	249	766	1,015	116.11	0.47
2020	-11,564	20,534	8,970	134.10	4.46
2021	-6,117	-1,389	-7,506	123.88	-3.47
2022	-757	3,169	2,412	111.23	0.99
2023	3,030	-9,469	-6,439	96.87	-2.38
2024	1,863	9,013	10,875	93.48	3.75
2025	2,059	4,161	6,219	89.67	2.03

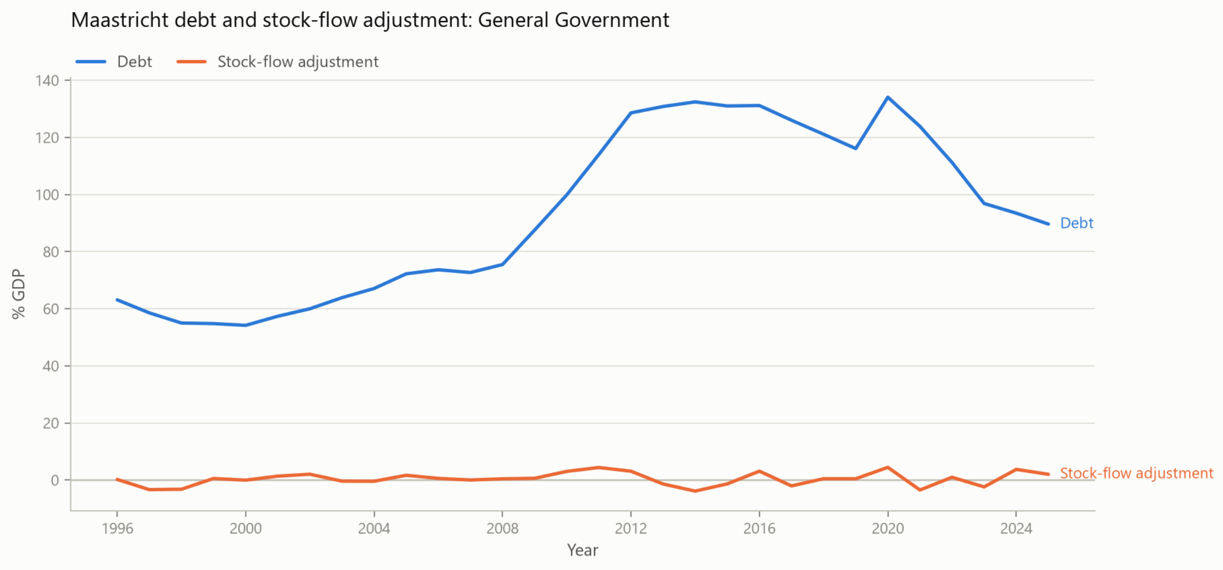

Figure 19: General Government Maastricht debt and stock-flow adjustment.

The stock-flow adjustment is a residual category, not a behavioural variable: a large value is a signal to consult the source documentation, not an anomaly in itself. The debt ratio also moves with nominal GDP, so a falling ratio does not imply falling debt.

C Intergovernmental Transfers

This appendix quantifies a convention rather than reporting a result. Because the subsectors are consolidated, a transfer between them changes the components and not the aggregate, so the recorded *location* of a balance depends partly on the transfer convention. That bears on how the composition results in the body should be read, which is why it is documented; it is not itself one of those results.

Historical source tables identify current and capital transfers received and paid between public administrations, which supports a mechanical sensitivity for 1977–1995:

$$B_{i,t}^{sens} = B_{i,t} - \left(T_{i,t}^{received} - T_{i,t}^{paid}\right).$$

The result is stored in `outputs/tables/historical_transfer_reallocation_sensitivity.csv`.

It is deliberately not labelled an underlying or true balance, and it is never used to restate B.9 anywhere in this repository. A transfer normally finances an expenditure responsibility assigned to the recipient, so removing the transfer while leaving the responsibility in place does not describe a coherent alternative arrangement. What the sensitivity quantifies is how much the recorded *location* of a balance depends on the transfer convention.

D Descriptive Macroeconomic Co-Movement

This material is placed in an appendix rather than the body because the specification is weak in ways that no amount of caveat wording repairs. It is retained because the estimates are part of the reproducible output, not because the report rests on them. Nothing in the body of this report depends on anything below.

The full-period specification is

$$B_{i,t}/GDP_t = \alpha_i + \beta_i g_t^{nominal} + \gamma_i I(t \geq 1995) + \epsilon_{i,t},$$

estimated with HAC standard errors. Nominal GDP growth is used because it is the one macroeconomic variable that can be reconstructed on a consistent basis from the bundled sources across the whole panel. It is deliberately not described as an output gap or as a cyclical adjustment, and the regime indicator is a statistical control for the 1995 splice rather than an estimate of a policy change.

D.1 Why these estimates carry little weight

Four specific problems, stated plainly.

1. **The dependent variable and the regressor share a construction.** The left-hand side carries nominal GDP in its denominator while the right-hand side is the growth rate of that same quantity. Part of any measured association is mechanical rather than economic, and the specification provides no way to separate the two parts.
2. **Nominal growth mixes two things.** With $g^{nominal} \approx g^{real} + \pi$, a single coefficient is asked to represent both real growth and inflation, which have no reason to move the balance by the same amount.
3. **The fit is very low and the coefficients are not significant.** The R^2 values span 0.07 to 0.18, and no nominal-growth coefficient reaches conventional significance under HAC standard errors.
4. **The labour specification rests on 18 observations.** At that sample size the positive unemployment coefficient below should not be interpreted at all: trends, collinearity between the labour series, dynamic specification and the time-series properties of the variables are all unexamined, and any one of them could account for the sign.

The contribution-base section of this report takes that narrower route for Social Security contributions, relating their change to the change in the aggregate employee wage bill $W_t = N_t \bar{w}_t$. That exercise is reported separately because it uses a quantity closer to the contribution base, while the specification below remains a weak aggregate co-movement diagnostic.

Table 32: Descriptive co-movement of subsector balance ratios with nominal GDP growth, HAC standard errors.

Sector	N	R^2	Growth coef.	HAC s.e.	HAC p	Regime coef.
General Government	48	0.1751	7.7012	8.8215	0.3827	3.6653
Central Government	48	0.1634	5.5312	7.6465	0.4695	2.8982
Regional and Local	48	0.0807	-0.0090	0.8348	0.9914	0.2213
Social Security Funds	48	0.0707	2.1789	2.0029	0.2766	0.5458

Table 33: Historical Social Security co-movement with the labour market, 1978–1995, over 18 annual observations.

Quantity	Estimate
R-squared	0.2494
Employment growth coefficient	9.7252
Employment growth, HAC standard error	5.2069
Employment growth, HAC p-value	0.0618
Unemployment rate coefficient	20.2847
Unemployment rate, HAC standard error	8.5997
Unemployment rate, HAC p-value	0.0183

These regressions quantify co-movement and nothing more. They are not causal estimates: no identification strategy is claimed, the samples are short and serially correlated, and HAC standard errors address the inference arithmetic rather than small-sample or specification risk.

E Artefact Index

Each section of this report is produced by a notebook and reads a persisted artefact.

Table 34: Report sections mapped to the notebook that produces them and the primary persisted artefact they read.

Section	Notebook	Primary artefact
Data, sources and validation	`03_harmonize_and_validate`	`data/processed/fiscal_balances_1977_2025.csv`
Identity and source-agreement checks	`03_harmonize_and_validate`	`outputs/tables/source_validation_summary.csv`
Long-run subsector decomposition	`04_balance_decomposition`	`data/processed/annual_balance_metrics_1977_2025.csv`
Year-to-year attribution	`06_year_to_year_attribution`	`outputs/tables/balance_change_attribution.csv`
Largest annual movements	`06_year_to_year_attribution`	`outputs/tables/largest_balance_movements.csv`
Revenue and expenditure dynamics	`05_revenue_expenditure`	`outputs/tables/revenue_expenditure_change_decomposition.csv`
Social Security Funds	`09_social_security_mechanisms`	`outputs/tables/social_security_account_metrics.csv`
Social Security balance-change decomposition	`09_social_security_mechanisms`	`outputs/tables/ssf_balance_change_decomposition.csv`
Social Security accounting boundaries	`09_social_security_mechanisms`	`outputs/tables/ssf_accounting_boundary_comparison.csv`
Primary balance and interest	`11_primary_balance`	`outputs/tables/primary_balance_and_interest.csv`
Primary balance sign frequencies	`11_primary_balance`	`outputs/tables/primary_balance_sign_summary.csv`
Fixed-capital-formation diagnostic	`12_investment_diagnostic`	`outputs/tables/investment_diagnostic.csv`
Debt and stock-flow adjustment	`13_debt_reconciliation`	`outputs/tables/debt_stock_flow_reconciliation.csv`
Persistence, pooled and by regime	`07_persistence`	`outputs/tables/persistence_by_regime.csv`
Structural mean shifts	`08_structural_breaks`	`outputs/tables/structural_breaks.csv`
Change-point robustness	`08_structural_breaks`	`outputs/tables/structural_break_sensitivity.csv`
Contribution base	`09_social_security_mechanisms`	`outputs/tables/contribution_change_decomposition.csv`
European benchmark	`16_european_benchmark`	`outputs/tables/european_benchmark_summary.csv`
Descriptive macroeconomic co-movement	`14_macroeconomic_comovement`	`outputs/tables/nominal_gdp_balance_comovement.csv`
Intergovernmental transfers	`10_intergovernmental_transfers`	`outputs/tables/historical_transfer_reallocation_sensitivity.csv`

Notebook names omit the `notebooks/` prefix and the `.ipynb` suffix. Figures are read from `outputs/figures/`.